3. Retrained the Core Scikit-Learn Models

Overview

With the new master_pbp_participation_2024.csv created and the target variables shifted to EPA, we had to execute a complete retraining of the digitalOC AI brains. The old models were purged, and new models were exported that natively understand advanced defensive coverages.

Code Changes

- **Execution:** Ran the updated Scikit-Learn training scripts (pbp_situation_model.py, exp_run_yards_model.py, exp_pass_yards_model.py) locally.
- **Deployment Integration:** Exported the newly trained brains as .joblib files and uploaded them to the project's GitHub Releases page. Updated the BASE_URL fetch variables in app.py to ensure the Flask server downloads these updated, coverage-aware models upon boot.

Future Roadmap

Right now, the models are utilizing Random Forest and Ridge Regression algorithms. Moving forward, we plan to experiment with XGBoost (Extreme Gradient Boosting) and deep neural networks (PyTorch) to see if we can extract even more predictive accuracy out of the highly dimensional pre-snap data.